

Ademy NLP, M1, P1
course

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Sec - 22
Inferential statistics

Sec - 15

Hypothesis testing in
ANOVA
(Partitioning of Variance
in ANOVA)

Null Hypotheses

$$H_0 :- \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

Alternate Hypothesis:

H_1 :- At least one of
are same

$$\mu_1 \neq \mu_2 \neq \mu_3 \dots \neq \mu_k$$

test statistics

$$F = \frac{\text{variance b/w samples}}{\text{variance within samples}}$$

variance b/w samples

	x_1	x_2	x_3
1	6	5	
2	9	6	
3	3	3	
4	5	2	2
5	3	1	4

variance within samples

The variance is given by σ^2

(3)

Standard Deviation is
is the distance or
deviation of a point
from the mean

x_1	x_2	x_3
1	6	9
2	7	6
4	3	3
5	2	2
3	1	4
$\bar{x}_1 = 3$	$\bar{x}_2 = \frac{19}{5}$	$\bar{x}_3 = 4$

$$H_0 = \bar{x}_1 = \bar{x}_2 = \bar{x}_3$$

H_1 = At least one sample
mean is not
equal.

6) on factor with
at least 2 levels are
independent.

(1) factors want to test
a machine which
reduces the feedback.

They split the
participants into 3
conditions [15 ms, 30 ms,
45 ms]

Later on, they asked
the participant to rate
the feedback [1-10]

Are there any differences
between the 3 conditions
using $\alpha = 0.05$?

(3)

SDs	15 mg	30 mg	45 mg
	9	7	4
	8	6	3
	7	6	2
	8	7	3
	8	8	4
	9	7	3
	8	6	2

(1) Define NULL & Alternate Hypothesis?

$$H_0: \mu_{15} = \mu_{30} = \mu_{45}$$

H_1 : not all were equal

(2) Significance $\alpha = 0.05$
 $C = 0.95$

(3) Calculate degree of Freedom

$$N = 21 \quad w = 3 \quad n = 1$$

$$\begin{aligned} \text{degree of freedom} &= w - 1 = 3 - 1 \\ &= 2 \\ &\quad (b/w) \end{aligned}$$

$$\begin{aligned} \text{df (within)} &= N - w \\ &= 21 - 3 = 18 \end{aligned}$$

df 11 df 12

(2, 18)



F table

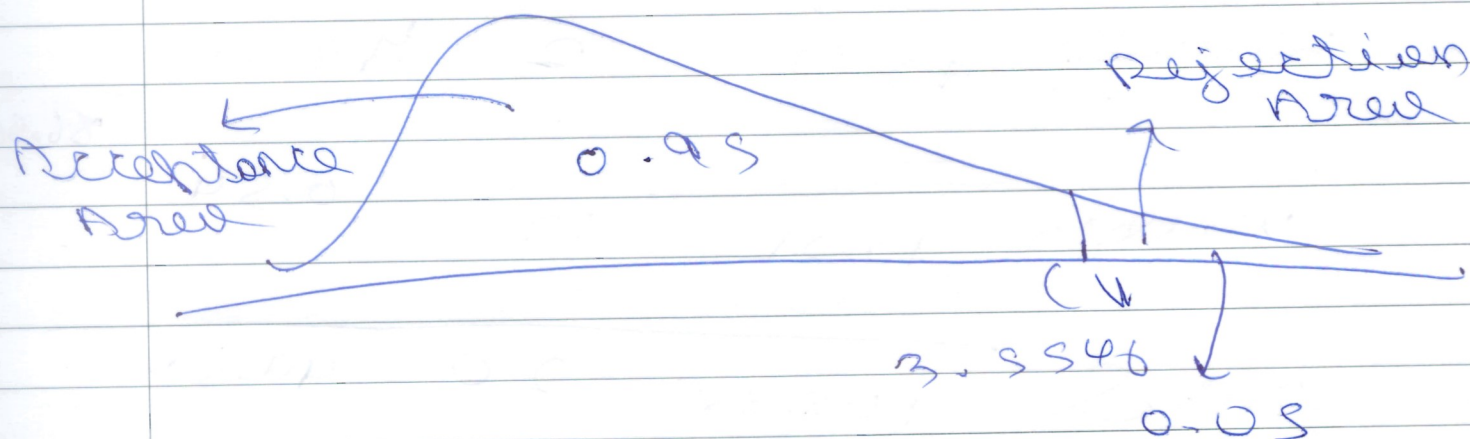
$$\alpha = 0.05$$



(initial value)

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Decision Boundary



① Decision rule

If F is greater than
 $3.5546 \rightarrow$ reject the
 Null Hypothesis.

② Calculate F test statistics

$$F = \frac{\text{variance b/w sample}}{\text{variance within sample}}$$

	SS	df	MS	F
Between	98.61	2	49.34	86.56
within	10.21	18	0.54	
total	108.88	20	49.88	

$$SS_{\text{between}} = \frac{\sum (\sum x_i)^2}{n} - \frac{T^2}{N}$$

→ For 15 mg :- 250 dose
 $\Rightarrow 4 + 8 + 7 + 8 + 8 + 9 + 8$

For 30 mg it will be $= 50$

30 mg : $7 + 6 + 6 + 7 + 8 + 7 + 6$
 $= 47$

45 mg : $4 + 3 + 3 + 8 + 4 + 3 + 2$
 $= 27$

(9)

$$= 1 \frac{572 \times 47^2 + 213}{9} - \frac{[572 + 47^2 + 21^2]}{21}$$

$$= 98.67$$

② $SS_{within} = \sum y^2 - \frac{\sum (\sum x_i)^2}{n}$

$$\sum y^2 = 9^2 + 8^2 + 7^2 + 8^2 + 8^2 + \dots$$

$$= 853$$

$$F_{test} = \frac{MS_{between}}{MS_{within}}$$

$$= \frac{49.34}{0.84}$$

$$= 86.56$$

2) F is greater than
3.5546, Reject the H_0 .

$$86.56 > 3.5546$$

$\therefore$ We reject null hypothesis.